

THE P-CURVE

Diagnosing Evidential
Value in a Literature

At a glance: The p-curve examines the distribution of statistically significant p-values (those between 0 and .05) across a set of studies. Genuine effects produce a right-skewed distribution — many p-values clustered near zero. A flat or left-skewed curve is a diagnostic signature of p-hacking or an absent true effect. The method requires only the test statistics from a set of significant results — no raw data needed.

1. Overview

The p-curve, introduced by Uri Simonsohn, Leif Nelson, and Joseph Simmons in 2014, is a tool for diagnosing the evidential value of a set of published findings. Its central insight is deceptively simple: if a true effect exists and studies are conducted with adequate power, most significant p-values will cluster near the low end of the 0–.05 range — not spread evenly across it, and not piled up just below .05.

When researchers p-hack — running many analyses and reporting only the one that crossed the threshold — they tend to produce p-values that are just barely significant: concentrated near .04 or .049 rather than near .001 or .01. This selective process leaves a distinctive fingerprint in the distribution of published p-values. The p-curve reads that fingerprint.

The method addresses a practical gap: where multiverse analysis requires access to raw data to diagnose fragility prospectively, p-curve can be applied retrospectively to an entire literature using only the test statistics reported in published papers. This makes it one of the most accessible meta-scientific tools for graduate students and researchers without data access.

2. Methodological Overview

The analyst collects the focal hypothesis test from each study in a set — the single test most directly tied to the paper's primary claim. These test statistics (t , F , z , χ^2 , r) are converted to exact p-values and plotted as a distribution across the 0–.05 range.

Two diagnostic tests are then applied:

Right-skew test: H_0 = uniform distribution across 0–.05

If the distribution is right-skewed (many p-values near 0), the test rejects H_0 — indicating evidential value. The test uses a half-normal approximation.

Flatness test: H_0 = 33% power distribution (pp-values)

Tests whether the p-curve is flatter than expected under studies with 33% power. A flat result relative to this benchmark indicates p-hacking without a true effect.

The 'pp-value' transformation is central to the method: each p-value is transformed into the probability of observing a p-value at least as small, given a specific alternative hypothesis. Under the null of no true effect combined with p-hacking, this transformed distribution is uniform; under a genuine effect, it skews right.

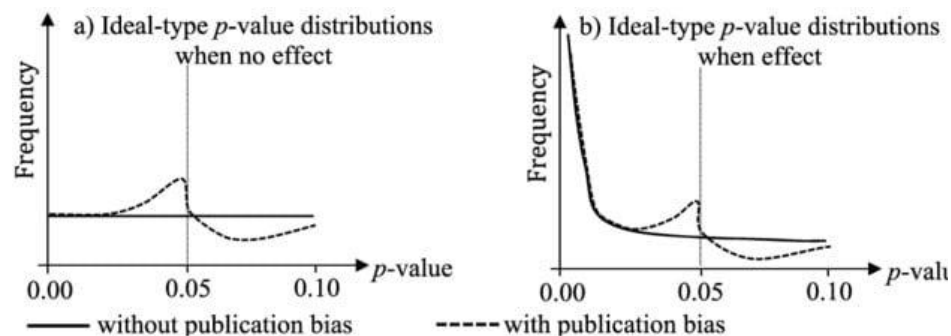
Simonsohn et al. recommend testing for both right-skew and flatness — a curve that is right-skewed passes the first test (evidential value exists) and must also pass the second test for the evidential value to be considered robust. A flat curve that fails the second test is consistent with widespread p-hacking that incidentally produces just-significant results.

The p-curve app (pcurve.com) automates all calculations. The analyst enters test statistics directly and the app produces the curve, the two tests, and an estimate of the average statistical power underlying the studies — a useful additional output when comparing literatures.

3. Interpreting the Curve Shape

Reading a p-curve is partly visual and partly inferential. Three canonical shapes matter:

Right-skewed	Many p-values near 0, few near .05. Signature of genuine effects with adequate power. The literature has evidential value — effects are real, though they may be smaller than the published estimates suggest if publication bias is also present.
Flat / uniform	p-values spread roughly evenly across 0–.05. Consistent with p-hacking in the absence of a true effect: researchers ran many tests and reported whichever crossed the line, producing a uniform distribution rather than the right-skew a real effect would generate.
Left-skewed	p-values piled up near .049. The strongest signature of p-hacking: studies are barely significant because researchers stopped as soon as they crossed the threshold. Reported effects almost certainly overstate or fabricate the underlying relationship. Also called a 'left-skewed p-curve'.



Hirschauer, Norbert & Grüner, Sven & Musshoff, Oliver & Becker, Claudia. (2018). Pitfalls of significance testing and $\$p$ -value variability: An econometrics perspective. *Statistics Surveys*. 12. 136-172. 10.1214/18-SS122.

4. Example Application: Ego Depletion

The ego depletion literature — the highly influential finding that acts of self-control deplete a limited mental resource — was a prominent target of p-curve analysis. Simonsohn and colleagues applied p-curve to the set of studies Hagger et al. (2010) used in a widely-cited meta-analysis supporting the effect.

The resulting p-curve was notably flat and partially left-skewed, failing the evidential value test. The analysis indicated that the published significant findings were consistent with p-hacking rather than a genuine underlying effect. This was a striking result because the individual studies had each passed peer review and the meta-analytic effect size appeared substantial.

The p-curve finding preceded by two years the large pre-registered multi-lab replication by Hagger et al. (2016) — across 23 labs — which found $d \approx 0.04$, essentially zero. The p-curve had correctly diagnosed the absence of evidential value from the published record alone, without requiring any new data collection.

5. Requirements & Applicability

The p-curve has specific requirements that govern when and how it can be applied appropriately.

Test statistics	The exact test statistic (t , F , z , χ^2 , r) and degrees of freedom for each study's focal test. These must be recoverable from the paper — p-values alone are insufficient because they cannot be back-transformed to exact statistics without df.
Focal test selection	One test per study — the test most directly linked to the primary theoretical claim. Multi-study papers should contribute one test per study, not one per

	paper. This is the most consequential analyst decision and must be documented.
Significant results only	The p-curve is computed using only significant results ($p < .05$). Non-significant results are excluded from the curve itself, though their presence is noted as context. This is by design — the method diagnoses the shape of the significant portion.
Minimum study count	Reliable inference requires approximately 20–25 studies. With fewer than 10 significant results the curve is too noisy for the formal tests, though visual inspection may still be informative.
Raw data required?	No. The method operates entirely on reported test statistics in published papers. This is p-curve's key practical advantage over multiverse and specification curve analysis.
Consistent hypothesis	All studies should be testing the same directional hypothesis. Mixing studies that test different effects or directions produces an uninterpretable aggregate curve.
No pooling of effects	p-curve is not a substitute for meta-analysis — it does not estimate the pooled effect size. It diagnoses evidential value, not magnitude. The p-curve app does report a power estimate, which implies an effect size, but this is secondary and should be interpreted cautiously.

6. Limitations

- p-curve diagnoses the significant portion of the literature only. It cannot speak to unpublished null results or the severity of publication bias — a right-skewed curve is consistent with genuine effects even in a severely publication-biased literature.
- The focal test selection is consequential and subjective. Different analysts may disagree on which test represents the primary claim of a paper, producing different curves from the same literature. Sensitivity analyses across reasonable alternative focal tests are recommended.
- The method assumes all studies test the same directional hypothesis with reasonable power. Mixing underpowered studies, different operationalizations, or different populations can distort the curve shape independently of p-hacking.
- p-curve cannot distinguish p-hacking from a true effect that is very small and the studies are severely underpowered — both produce flat curves. The flatness test tests against a 33% power benchmark, which may be too high or too low for some literatures.
- The power estimate derived from the p-curve rests on several assumptions and should not be treated as a precise estimate. It provides a rough diagnostic, not a substitute for a formal power analysis.
- p-curve applies only within the 0–.05 significance range. Results reported as exact p-values below .001 are often truncated in papers (reported as ' $p < .001$ ') and must be entered conservatively, which can affect the curve shape.

7. How to Apply

Using the p-curve app (pcurve.com)

The web app maintained by Simonsohn, Nelson, and Simmons is the standard implementation. It accepts test statistics entered in a simple text format (e.g., ' $t(38) = 2.31$ ' or ' $F(1,76) = 5.41$ ') and produces:

- The p-curve plot with the two diagnostic test results
- A statistical power estimate for the studies in the set
- Downloadable results for reporting

Step-by-step procedure

- Identify the set of studies: define inclusion criteria before examining the results — typically all direct tests of the focal hypothesis in the literature.

- For each study, identify the single focal test — the statistical test most directly tied to the primary theoretical claim. Document the selection rule explicitly.
- Collect the exact test statistic and degrees of freedom from each paper. If a paper reports only $p < .001$, enter as $p = .001$ (conservative). Exclude non-significant results.
- Enter each statistic into the p-curve app and run the two tests. Report both the right-skew test and the flatness test, along with the visual curve.
- Conduct sensitivity analyses: re-run the curve with plausible alternative focal test selections to check whether conclusions are robust to this choice.

R implementation

The dmetar R package (Harrer et al. 2021) includes a `pcurve()` function that accepts a vector of p-values and produces the curve and tests. This is useful for batch processing of large literature sets and for integration with meta-analytic workflows.

8. Related Tools

p-uniform	A complementary p-value distribution method that also operates on significant p-values but models the full distribution under a specific assumed effect size. More sensitive to heterogeneity than p-curve; the two methods should converge when both assumptions are met.
Caliper test	Tests for an excess of results just below the $p = .05$ threshold relative to just above it — a direct signature of stopping-rule manipulation or rounding. Less powerful than p-curve but simpler and requires only a count of results above and below the threshold.
z-curve 2.0	Extends the p-value distribution approach to include non-significant results, estimating the expected replication rate and discovery rate across the full distribution of test statistics. Complements p-curve by working with the complete set rather than only the significant portion.
Excess significance test (TES)	Compares the observed count of significant results to the count expected given study power. Where p-curve diagnoses the shape of the significant distribution, TES diagnoses whether there are too many significant results in the first place.
Funnel plot / Egger test	Meta-analytic tools that detect publication bias by examining asymmetry in effect size vs. precision plots. Complementary to p-curve: funnel asymmetry reflects missing studies; a left-skewed p-curve reflects within-study manipulation.

Key Citations

- Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). P-curve: A key to the file drawer. *Journal of Experimental Psychology: General*, 143(2), 534–547. <https://doi.org/10.1037/a0033242>
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